

Vrica Gede Penggalih

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Summary

Software Engineer with hands-on experience in backend and frontend development and low-level engineering, specializing in building web applications and system-level projects. Experienced in designing RESTful APIs, implementing authentication and authorization systems (JWT, RBAC), and working with relational (SQL) and NoSQL databases.

Flexible engineer comfortable working with various tech stacks, from high-level web frameworks like React and Laravel to raw socket network programming and compiler development.

Technical Skills

- Languages: Go, Javascript, Python, PHP, Shell
- Web Technologies: React, Laravel, NodeJS (ExpressJS)
- Database: MongoDB, PostgreSQL, MySQL, Redis
- Concepts: REST API, JWT Authentication, RBAC, HTTP Protocol, Monolithic Architecture
- Tools: Git, Linux (Arch Linux), Docker, AWS, Postman, CPanel

Experience

Product Intern **PT Aino Teknologi** **Oct 2024 – Dec 2024**

- Developed an internal ERP system (Project Dashboard) using Laravel, PostgreSQL, Spatie Permission, and Laravel Excel.
- Designed and implemented Role-Based Access Control (RBAC) with Admin, Project Manager, and Finance roles.
- Designed and developed the Frontend, the Backend, the Database schema and the RESTful APIs.
- Contributed to improving internal project tracking and management.

Teaching Programming (proker KKN) **SMK Muhammadiyah 1 Playen** **August 2025**

- Taught basic programming to vocational high school students (TKJ major).
- Used Bahasa Indonesia-based programming language (BAIK) to help students understand programming logic more intuitively.
- Designed structured learning materials to accelerate students' understanding of core programming concepts.

Selected Projects

Kusmala – Interpreted Programming Language **Sept 2024 – Dec 2024**

Live Demo: <https://ku-7a61eb073c154657985fb0a5e6bce6e6.ecs.ap-southeast-1.on.aws/>

- Developed my own interpreted programming language with Indonesian-based syntax.
- Built lexer, parser, AST, and interpreter in Go.
- Designed language grammar and execution model from scratch.
- Built live/online kusmala playground for an easy online usage.

Gherver – HTTP Server library from Scratch **May 2025**

- Built my own HTTP library from scratch in Go.
- Implemented TCP socket handling, HTTP request parsing, and response handling from the ground up.
- Designed a simple routing system supporting HTTP methods and custom handlers.
- Added JSON and static file serving for documents and media.
- Deepened understanding of HTTP protocol and low-level networking concepts.

DMS AINO – Document Management System **Sept 2025 – Dec 2025**

Live Demo: dms-aino.onrender.com

- Developed a Enterprise web-based Document Management System for PT AINO Indonesia as part of Capstone Project in Uni.
- Role as the only programmer (full stack) in the group.
- Support Upload, View, Send, Sign, Audit, Download, Filtering and Delete document.
- Implemented sequential, parallel, and group-based digital signing workflows.
- Designed and developed the frontend (React), the backend (Node.js + Express) and the database (MongoDB + Mongoose).
- Implemented JWT authentication and RBAC authorization.
- Implemented RESTful API architecture in a monolithic system and managing the deployment.

Education

- **S1 in Information Technology**, Universitas Aisyiyah Yogyakarta (gpa 3.66). **2022 – 2026**
Thesis: Comparative Analysis Of YOLO Model Performance For Freshwater Fish Detection, JIPI journal, SINTA 3

Certificate

- Machine Learning – Dicoding (Final Project: 4/5) **Feb 2024**
- Backend Developer – Dicoding (Final Project: 5/5) **Sept 2024**
- AWS cloud – Dicoding **Sept 2024**